

1.Basic Prompt writing: Create two different prompts to ask an AI about the topic "Electricity." The first prompt should be vague, and the second prompt should be clear and specific. Compare the responses you get and describe which prompt gave a better answer and why.

1.1Tell Me About Electricity:

Response:

- Electricity is a form of energy caused by the movement of **electrons**.
- It flows through materials called **conductors** (like copper wires).
- The flow of electricity is called **electric current**.
- **Voltage** pushes the current, and **resistance** opposes it.
- There are two types: **static electricity** and **current electricity**.
- Electric current is of two types: **AC (Alternating Current)** and **DC (Direct Current)**.
- Electricity is generated from sources like **solar, wind, water, and thermal power**.
- It is used for lighting, running appliances, industries, and communication.
- It is essential for modern life and technology.
- Proper safety measures are important to avoid electrical accidents.

1.2Clear and specific information of electricity (types of electricity and its description)

Response:

Electricity is mainly divided into two types: **static electricity** and **current electricity**. Static electricity is the accumulation of electric charges on the surface of an object. These charges remain at rest and do not flow continuously. It is usually produced by friction when two objects are rubbed together, causing electrons to transfer from one object to another. Common examples include a balloon sticking to a wall after rubbing it on hair and the occurrence of lightning.

Current electricity, on the other hand, is the continuous flow of electric charges through a conductor such as a wire. This type of electricity is used in homes, industries, and electronic devices. It requires a closed circuit to allow the flow of electrons. Current electricity is further classified into two types: direct current (DC) and alternating current (AC). Direct current flows in one direction and is commonly used in batteries and electronic devices. Alternating current changes its direction periodically and is used for household power supply and large-scale electricity distribution.

Static Electricity

Static electricity occurs when electric charges accumulate on the surface of an object and remain at rest. It is usually produced due to **friction**, when electrons transfer from one material to another. As a result, one object becomes negatively charged and the other positively charged. These charges stay in one place until they are released suddenly, often causing a small spark or shock. A common natural example is **lightning**,

Current Electricity

Current electricity is the continuous flow of electric charges through a conductor, such as a metal wire. It is the most useful type of electricity and is widely used in daily life. For current to flow, a **closed circuit** is required, along with a source like a battery or generator. This type of electricity powers homes, industries, and electronic devices.

Current electricity is further divided into two types:

- **Direct Current (DC):** Flows in only one direction. It is commonly produced by batteries, cells, and solar panels, and is used in devices like mobile phones and laptops.
- **Alternating Current (AC):** Changes direction periodically. It is generated in power stations and supplied to homes and industries because it can travel long distances efficiently.

CONCLUSION:

In conclusion, both prompts were about electricity, but the clear and specific prompt gave a better answer than the simple prompt. As we can see that simple prompt provide us just a basic information about the electricity but mean while if give the prompt on a specific task we see clearly that it gives in detail. Like prompt on electricity generation, different types of electricity, generators, and different energy sources clearly. This shows that using plain but specific words helps Chat GPT understand the requirement and give accurate, detailed, and useful responses.

2. Zero-Shot Prompting: Create a prompt that asks an AI to explain Ohm's Law without giving any example or background. Evaluate how well the AI explains the concept based on your prompt alone.

Prompt: Explain Ohm's Law

Ohm's Law states that the **current (I)** flowing through a conductor is directly proportional to the **voltage (V)** across it and inversely proportional to its **resistance (R)**, provided the physical conditions like temperature remain constant.

Formula: $V = IR$

Where:

- **V (Voltage):** Potential difference applied across the conductor (Volts)
- **I (Current):** Flow of electric charge (Amperes)
- **R (Resistance):** Opposition to current flow (Ohms, Ω)

Ohm's Law helps us understand how electricity behaves in a circuit:

- When **voltage increases**, it pushes more electrons, so **current increases**.
- When **resistance increases**, it restricts electron flow, so **current decreases**.
- If resistance is constant, voltage and current have a **linear relationship**.

This means the graph between voltage and current is a **straight line passing through the origin**.

Evaluation:

The AI explained the concept clearly without any example or background information the response was short, accurate, and easy to understand. However, adding examples or applications would make the explanation better and more practical.

3. One-Shot and Few-Shot Prompting: Provide the AI with a single example of how to calculate the resistance in a simple circuit. Then write your own prompt asking the AI to solve a similar resistance calculation. After that, add two more examples to your prompt and observe any changes in the AI's response quality.

One-Shot Prompt:

Example:

Voltage (V) = 12 V, Current (I) = 3 A Using Ohm's Law: $R = V / I = 12 / 3 = 4 \Omega$

Now solve:

Calculate the resistance in a circuit where the voltage is 20 V and the current is 4 A using Ohm's Law."

AI Response:

Resistance= $V/I = 20/5 = 4 \Omega$.

Few-Shot Prompt:

Example 1: $V = 12 \text{ V}, I = 3 \text{ A} \rightarrow R = 4 \Omega$

Example 2: $V = 10 \text{ V}, I = 2 \text{ A} \rightarrow R = 5 \Omega$

Example 3: $V = 24 \text{ V}, I = 6 \text{ A} \rightarrow R = 4 \Omega$

Now calculate the resistance when voltage is 18 V and current is 3 A.

AI Response:

Resistance= $V/I = 18 / 3 = 6 \Omega$

Conclusion:

With **one example**, AI gives a correct but basic answer. With more examples, the AI better understood the pattern and gave a faster, clearer response. Few-shot prompting improves accuracy and response quality.

4. Chain-of-Thought Prompting: Develop a prompt that guides the AI step-by-step through calculating current flow in a circuit using Ohm's Law with resistors in series. Then, ask a final question for the AI to solve. Analyse how breaking down the reasoning steps impacts the accuracy of the answer.

Prompt:

"You are given a simple electrical circuit with resistors connected in series. Follow these steps to calculate the current:

AI Response:

Step 1: Identify the given values: voltage (V) and resistances (R_1 , R_2 , R_3).

Step 2: Calculate the total resistance in series using: $R_{\text{total}} = R_1 + R_2 + R_3$

Step 3: Apply Ohm's Law: $I = V / R_{\text{total}}$

Step 4: Substitute the values and calculate the current step by step.

- Given: $V = 24 \text{ V}$, $R_1 = 2 \text{ } \Omega$, $R_2 = 4 \text{ } \Omega$, $R_3 = 6 \text{ } \Omega$
- Total Resistance: $R_{\text{total}} = 2 + 4 + 6 = \mathbf{12 \text{ } \Omega}$
- Using Ohm's Law: $I = V / R_{\text{total}} = 24 / 12$
- Current: $I = \mathbf{2 \text{ A}}$

Conclusion:

Breaking the problem into steps helped the AI give a clear and accurate answer. Chain-of-thought prompting improves understanding, reduces mistakes, and makes the solution easy

5) Prompt Refinement: Start with an ambiguous prompt related to the "Water Cycle." Test the AI's response, note the confusion or errors, and then refine your prompt to make it clearer and more specific. Repeat this process twice and record how the AI's responses improve with each refinement.

Role-Based Prompting: Create three prompts asking the AI to explain "Newton's Laws of Motion," each with a different role instruction:

(a) as an expert engineer

(b) as a high school teacher

(c) as a beginner Compare the tone, detail, and style of the responses

First Prompt (Ambiguous): Explain the Water Cycle

AI Response:

The water cycle is the movement of water between Earth and the atmosphere.

Issue:

The answer was too short and lacked details.

Refined Prompt 1:

Explain the Water Cycle with its main stages.

AI Response:

The water cycle includes evaporation, condensation, precipitation, and collection. Water changes form and moves through nature continuously.

Refined Prompt 2:

Explain the Water Cycle in simple words with all stages and examples for students.

AI Response:

The water cycle is the continuous movement of water in the environment. It begins with **evaporation**, where water from oceans, rivers, and lakes turns into vapor due to heat from the sun. This vapor rises and cools to form clouds in a process called **condensation**. When the clouds become heavy, water falls back

to Earth as **precipitation**, such as rain, snow, or hail. The water then **collects** in rivers, lakes, and oceans, and the cycle repeats.

For example, rainwater fills rivers and oceans, which again evaporate due to sunlight. The water cycle is important because it maintains the balance of water on Earth, supports plant and animal life, and regulates climate.

CONCLUSION:

With each refinement, the AI response became clearer, more detailed, and easier to understand. Clear prompts give better results.

ROLE BASED PROMPTING:

Prompt 1 (Expert Engineer): Explain Newton's Laws of Motion as an expert engineer.

AI Response:

From an engineering perspective, **Newton's Laws of Motion** form the fundamental basis for analysing and designing mechanical systems. These laws describe how forces interact with physical bodies and are essential in fields such as structural, mechanical, and automotive engineering.

Prompt 2 (High School Teacher): Explain Newton's Laws of Motion as a high school teacher.

AI Response:

First Law (Law of Inertia): An object will stay at rest or keep moving in a straight line unless a force act on it.

Second Law (Force and Acceleration): The acceleration of an object depends on the force applied and its mass. Formula **$F=ma$**

Third Law (Action and Reaction): For every action, there is an equal and opposite reaction

Prompt 3 (Beginner): Explain Newton's Laws of Motion for a beginner.

AI Response:

Newton's laws tell us how things move. First, things do not move unless pushed. Second, heavier things need more force to move fast. Third, if you push something, it pushes back.

Comparison:

The expert response was technical and detailed, the teacher response was clear with examples, and the beginner response was very simple and easy to understand.

6) Creative Engineering Problem Prompts: Craft a prompt that asks the AI to brainstorm ideas for designing a low-cost water purification system suitable for rural areas. Encourage creativity by adding phrases like “limited resources” and “sustainability.”

Prompt: Act as a creative engineer and brainstorm innovative ideas for designing a low-cost water purification system suitable for rural areas. Consider conditions with limited resources, minimal electricity, and locally available materials. Focus on sustainability, ease of maintenance, and affordability. Suggest multiple design approaches, explain how each works, and include practical examples of materials that can be used.”

AI Response:

A low-cost water purification system suitable for rural areas can be effectively designed by combining simple filtration, natural treatment methods, and basic disinfection techniques using locally available materials. The process can begin with a **pre-filtration stage**, where dirty water is passed through a clean cloth to remove large particles like leaves, dust, and debris. This is followed by a **multi-layer filtration system** constructed using a plastic drum or clay pot. Inside the container, layers of gravel, coarse sand, fine sand, and charcoal (made from burnt wood or coconut shells) are arranged. As water flows through these layers, gravel removes larger impurities, sand filters finer particles, and charcoal helps eliminate odor, color, and some harmful chemicals.

After filtration, the water can be further purified using **solar disinfection (SODIS)**, where it is stored in transparent plastic bottles and exposed to direct sunlight for 6–8 hours. The ultraviolet rays from the sun help kill bacteria, viruses, and other microorganisms. Additionally, natural coagulants like **crushed moringa seeds** can be used before filtration to settle suspended particles and improve clarity. For maximum safety, especially in highly contaminated water, the filtered water can be **boiled using firewood or biomass**, which ensures complete destruction of harmful pathogens.

This integrated system is highly suitable for rural settings because it requires **minimal electricity, low cost, and simple maintenance**. All materials are easily available locally, making it affordable and sustainable. Moreover, the system can be easily understood and operated by local communities, ensuring long-term usability and access to safe drinking water.

7. Ethical Prompt Design Discussion: Identify a biased prompt related to job descriptions (e.g. language with respect to a gender). Rewrite the prompt to remove bias and create a neutral, inclusive version. Explain why this revision is more ethical.

1. Biased Prompt: Prompt: “Write a job description for a salesman who can handle customers aggressively and work long hours.”

AI Response:

We are looking for a hardworking salesman to join our company. He should be able to convince customers aggressively, work under pressure, and stay for long working hours. He must increase sales and manage customer needs.

2. Neutral & Inclusive Prompt: Prompt: “Write a job description for a sales executive who can communicate effectively with customers, solve problems professionally, and manage work responsibilities efficiently.”

AI Response:

We are hiring a sales executive to join our team. The candidate should have good communication skills, assist customers professionally, solve customer issues, and manage daily tasks efficiently. The role includes achieving sales targets, maintaining positive customer relationships, and supporting business growth.

3. Why Neutral Prompt is Better:

The first prompt is biased because it uses gender-specific language (“salesman”) and promotes aggressive behaviour. The second prompt is fair and inclusive because it uses gender-neutral language (“sales executive”) and focuses on communication, professionalism, and equal opportunity for all candidates.

8. Simulated Customer Support Chatbot: Develop a prompt that instructs the AI to play the role of a technical support agent helping a customer troubleshoot a failure in an electronic circuit. Include instructions to keep the tone friendly and professional and to ask diagnostic questions.

Prompt: “Act as a technical support chatbot helping a customer troubleshoot a failure in an electronic circuit. Keep the tone friendly and professional. Ask diagnostic questions step-by-step to identify the issue, suggest possible causes, and provide clear solutions.”

AI Response:

Hello! I’m here to help you troubleshoot your electronic circuit issue. Let’s solve it step-by-step.

- 1)Can you describe the problem? (For example: circuit not turning on, LED not glowing, motor not running, etc.)
- 2)Is the power supply connected properly and providing the correct voltage?
- 3)Are all wires and components connected correctly according to the circuit diagram?
- 4)Have you checked for loose connections or damaged wires?
- 5)Are any components heating up, burnt, or showing visible damage?
- 6)Have you tested the battery or power source recently?

Possible Causes:

- No power supply
- Loose wiring
- Wrong component connection
 - Damaged resistor, transistor, or IC
- Short circuit

Suggested Solutions:

- Reconnect all wires properly
- Replace weak battery or faulty power supply
- Check circuit diagram carefully

9. Multi-Language Prompting: Develop a prompt that asks the AI to translate a simple engineering glossary (5 technical terms) from English to your native language. Then modify the prompt to request additional explanations of these terms in the translated language. Multi-Language

Multi-Language Prompting

Initial Prompt:

Translate the following 5 engineering terms from English to Hindi:

- 1)Voltage
- 2)Current
- 3)Resistance
- 4)Circuit
- 5)Transformer

AI Response:

- 1)Voltage – वोल्टेज
- 2)Current – धारा
- 3)Resistance – प्रतिरोध
- 4)Circuit – परपथ
- 5)Transformer – परिवर्तित्र

Modified Prompt:

Translate the following 5 engineering terms from English to Hindi and give a simple explanation of each term in Hindi:

- 1)Voltage
- 2)Current
- 3)Resistance
- 4)Circuit
- 5)Transformer

AI Response:

1)Voltage – वोल्टेज

वोल्टेज वह बल है जो इलेक्ट्रॉनों को एक स्थान से दूसरे स्थान तक धकेलता है। इसे विद्युत का “दबाव” भी कहा जाता है।

2)Current – धारा

सरलव्याख्या:

करंट (धारा) इलेक्ट्रॉनों का प्रवाह है जो तार के माध्यम से बहता है। यह बताता है कि कितनी बिजली बह रही है।

3)Resistance – प्रतिरोध

सरलव्याख्या:

प्रतिरोध वह गुण है जो विद्युत धारा के प्रवाह को रोकता या कम करता है। अधिक प्रतिरोध होने पर धारा कम बहती है।

4)Circuit – पररपथ

सरलव्याख्या:

परिपथ एक बंद रास्ता होता है जिसमें से बिजली बहती है। यदि परिपथ पूरा नहीं है, तो धारा नहीं बहेगी।

5)Transformer – ट्रांसफॉर्मर

सरलव्याख्या:

ट्रांसफॉर्मर एक उपकरण है जो वोल्टेज को बढ़ाता या घटाता है। इसका उपयोग बिजली को लंबी दूरी तक भेजने में किया जाता है।

10. Review a curated set of different prompt types (e.g., for summarization, information extraction, paraphrasing, question answering) from a "Prompt Gallery." For each prompt type, match it with a real-world task (e.g., summarizing a lecture note, extracting names from a project report). Test at least three prompt templates on an AI tool or by role-play (students simulate being the AI), with varied wording. Record the outcomes and discuss which prompt (or template) was most effective for each task, and explain why you think it worked best. Reflect on how changing small parts of a prompt can alter model response quality, completeness, or accuracy.

Prompt Type	Description	Real-World Task
Summarization	Condenses long text into key points	Summarizing lecture notes or articles
Information Extraction	Pulls specific data (names, dates, etc.)	Extracting names from a project report
Paraphrasing	Rewrites text in a new way	Simplifying technical content
Question Answering (QA)	Answers questions based on given text	Answering exam questions from study material

These prompt types are commonly used in AI systems to improve productivity and clarity in tasks like education and data processing

Lecture notes: Summarizing a **lecture note on Electricity**

2. Testing Prompt Templates (3 Tasks)

Task 1: Summarization (Lecture Notes)

Template A

“Summarize the following text.”

Outcome:

- Very general summary
- Missed key points

Template B

“Summarize the following lecture note in 5 bullet points.”

Outcome:

- Clear structure
- Important ideas included

Template C

“Summarize the text in 5 bullet points, highlighting key concepts for beginners.”

Outcome:

- Best clarity
- Easy to understand
- Covered all key ideas

Most Effective: Template C

Reason: Adding constraints (bullet points + audience) improves completeness and clarity

Task 2: Information Extraction (Project Report)

Template A

“Extract important information from the text.”

Outcome:

- Vague output
- Missed specific details

Template B

“Extract all names and dates from the text.”

Outcome:

- More accurate

- Focused extraction

Template C

“Extract names and dates and present them in a table format.”

Outcome:

- Highly structured
- Easy to read and use

Most Effective:

Template C

Reason:

Structured formats (table/JSON) improve usability and accuracy

Task 3: Paraphrasing (Simplifying Text)

Template A

“Rewrite the sentence.”

Outcome:

- Minimal change
- Not very helpful

Template B

“Paraphrase the sentence in simple language.”

Outcome:

- Easier to understand
- Good clarity

Template C

“Paraphrase the text in simple language for a 10th-grade student while keeping the meaning.”

Outcome:

- Very clear
- Maintained meaning
- Improved readability

Most Effective: Template C

Reason: Specifying audience + clarity improves quality and usefulness

3. Overall Observations

- Prompts with clear instructions perform better than vague ones
- Adding format (bullet points, table) improves structure
- Including audience or purpose improves relevance
- Using specific verbs (extract, list, highlight) gives better results than generic words like “summarize”

4. Reflection on Prompt Changes Small changes in prompts can significantly affect output:

Example:

- “Summarize this text” → general summary
- “Summarize in bullet points with key concepts” → structured + complete

Key Effects:

- Quality: More detailed instructions = better answers
- Completeness: “Highlight key points” ensures nothing important is missed
- Accuracy: Specific extraction prompts reduce errors
- Clarity: Audience-based prompts simplify understanding

Even minor wording changes (like “bullet points” vs “paragraph”) can change the entire output format and usefulness

5. Conclusion

From testing different prompt templates:

- Best prompts are specific, structured, and goal-oriented
- Templates that include constraints + format + audience give the most effective results
- Iterative refinement of prompts is essential for improving AI performance

11. Choose a real engineering challenge or societal problem relevant to your field* (e.g., “Reducing plastic waste in campus cafeterias” or “Optimizing solar panel placement on campus rooftops”). Draft an initial prompt that asks an AI to propose practical solutions. Share the AI’s (or peer’s) answer in small groups and identify aspects that are missing, vague, or not actionable. Refine your prompt based on feedback (e.g., specify constraints, ask for step by-step solutions, or require a list of pros and cons). Repeat the process one more time, refining again for further clarity or specificity. Document the entire prompt-refinement process and share the best solution generated, along with a brief analysis of how prompt improvements led to better responses.

1. Chosen Problem

Optimizing solar panel placement on campus rooftops

This is an important engineering challenge because it improves renewable energy use, reduces electricity costs, and supports sustainability.

2. Initial Prompt (Version

1) Prompt:

“Suggest ways to optimize solar panel placement on campus rooftops.”

AI Response (Summary)

- Place panels on rooftops
- Use sunlight efficiently
- Avoid shaded areas
- Maintain panels regularly

Issues Identified

- Too **general and vague**
- No technical details
- No step-by-step plan
- No constraints (budget, area, etc.)
- Not actionable for real implementation

3. Refined Prompt (Version 2)

Prompt:

“Suggest practical ways to optimize solar panel placement on campus rooftops. Include factors like sunlight direction, shading, and panel angle. Provide step-by-step recommendations.”

AI Response (Summary)

- Analyse roof orientation
 - Tilt panels for maximum sunlight
 - Avoid shading from trees/buildings
 - Use proper spacing
 - Perform site survey

Improvements

- More **structured and relevant**
- Includes important factors
- Some actionable steps

Still Missing

- No cost considerations
- No comparison of options
- No real-world constraints (budget, maintenance, energy output)

4. Final Refined Prompt

(Version 3) Prompt:

“Propose a detailed and practical plan to optimize solar panel placement on campus rooftops. Include:

- Step-by-step implementation
- Key factors (orientation, tilt angle, shading, roof area)
- Estimated cost considerations
- Expected benefits (energy output, savings)
- Pros and cons of different placement strategies
- Recommendations for maintenance and monitoring”

AI Response (Best Output Summary)

- Conduct site survey using solar mapping tools
- Install panels facing south (in Northern Hemisphere) with optimal tilt angle
- Use spacing to avoid shading losses
- Compare fixed vs adjustable mounting systems
- Estimate installation cost and ROI
- Monitor performance using smart systems
- Provide maintenance schedule

Strengths

- **Highly detailed and actionable**
- Includes **engineering considerations**
- Covers cost, benefits, and trade-offs
- Suitable for real-world implementation

5.Final Best Solution (Consolidated)

A well-optimized solar panel system on campus rooftops should begin with a detailed site assessment using solar mapping tools to evaluate sunlight availability and shading. Panels should be oriented southward with an optimal tilt angle based on geographic location to maximize energy generation. Proper spacing must be maintained to avoid shadow overlap. Different mounting systems (fixed or adjustable) should be evaluated based on cost and efficiency.

The plan should also include cost estimation, expected energy output, and return on investment. Regular maintenance and monitoring systems should be implemented to ensure long-term efficiency. Using smart monitoring tools can help detect faults and improve performance over time.

6.Analysis of Prompt

Refinement Key Learnings:

1. **Adding specificity improves quality**
 - Vague prompt → generic answer

- Detailed prompt → actionable solution
- 2. Structured instructions improve clarity
 - Bullet points, step-by-step → organized output
- 3. Requesting comparisons improves depth
 - Pros & cons → better decision-making

CONCLUSION:

Small changes in prompts—such as adding “**step-by-step**,” “**cost**,” “**pros and cons**,” and “**constraints**”—greatly improved the AI’s response. The final prompt produced a solution that is not only clear but also practical and ready for implementation.

